

Candidate Name: _____ Class: 07S_____



JURONG JUNIOR COLLEGE

PRELIMINARY EXAMINATION 2008

H2 BIOLOGY 9747/2

FRIDAY

15 AUGUST 2008

2 HOURS

INSTRUCTIONS TO CANDIDATES :

Write your name and class in the spaces at the top of this page and on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Write your answer in the spaces provided on the question paper.

Section B

Answer any **one** question.

At the end of the examination

1. fasten any separate answer paper used securely to the question paper.
2. enter the question number from Section B in the grid opposite.

INFORMATION FOR CANDIDATES

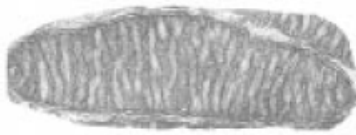
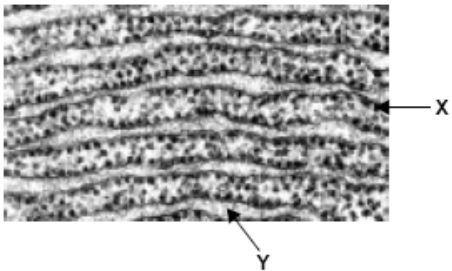
The number of marks is given in brackets [] at the end of each question or part question.

EXAMINER'S USE ONLY	
Section A	Mark
1	
2	
3	
4	
5	
6	
7	
8	
Section B	
Total	

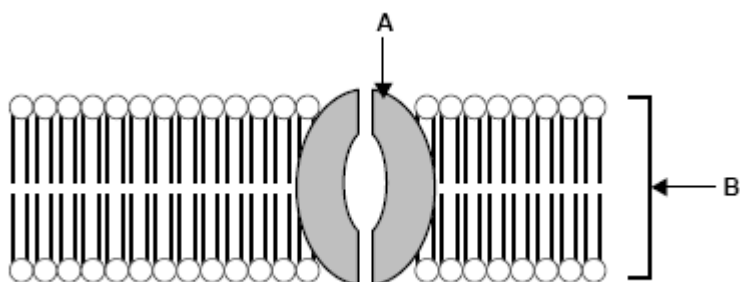
Section A

Answer **all** the questions in this section.

1. (a) The following table contains images of a number of structures seen in cells. Name each of the structures and outline the function of each. [5]

Structure		Name and function
	i.	
	ii.	Structure X
	iii.	Structure Y
	iv.	
	v.	

The diagram below represents a cross section of part of a cell membrane.



- (b) Name the structures labelled **A** and **B**.

Structure **A** _____ Structure **B** _____ [2]

The concentration of sodium ions, Na⁺, in human blood plasma is approximately 150 mmol/L. In the cytosol of red blood cells the concentration of these ions is between 25 and 30 mmol/L.

(c) Explain how this difference in concentration is maintained. [2]

Animal cells placed in distilled water swell and burst. Plant cells placed in distilled water do not burst.

(d) Explain why plant cells do not burst in this situation. [1]

[Total: 10]

2. During digestion, large molecules are broken down into their constituent monomers (subunits).

(a) Complete the table below. [2]

Large molecules	Monomers (subunits)
	Amino acids
Lipid	
	Monosaccharide
DNA	

(b) Explain how a monosaccharide enters an epithelial cell. [2]

(c) For what purpose would a cell use a lipid monomer? [1]

Starch, a storage molecule in plants, is a polymer made of glucose molecules joined together. It consists of two components.

(d) Name these two components. [2]

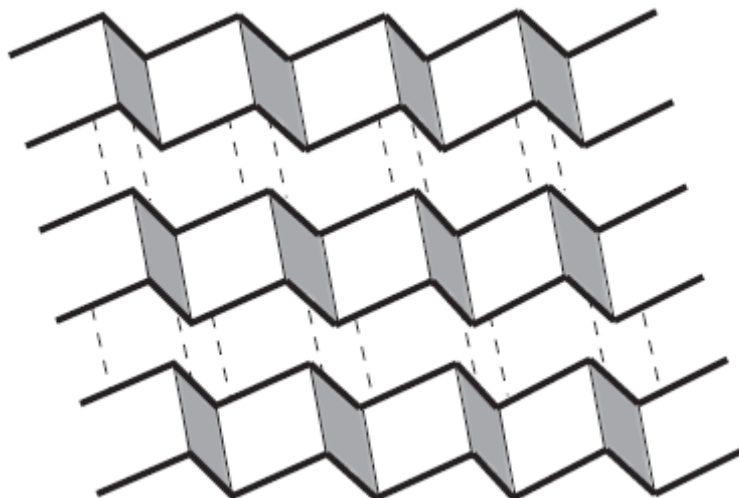
(e) (i) Give three similarities between these two components. [3]

(ii) Give two differences between these two components. [2]

Scientists are now turning to the study of the proteome (all of the proteins) of an organism rather than the study of single proteins.

(f) Suggest **one** reason why the emphasis is now on the study of all the proteins of an organism rather than on one protein at a time. [1]

Protein molecules come in many shapes and forms that can be classified into primary, secondary, tertiary and quaternary. The secondary, tertiary and quaternary shapes arise as a result of different kinds of folding of a primary structure. One kind of secondary structure is a pleated sheet where the primary molecule extends along the folded sheet. The primary structures in the layers are held together by hydrogen bonding.



(g) Suggest two ways in which such a structure may be important to the functioning of the protein. [2]

[Total: 15]

3. In *Drosophila* flies, a gene for wing shape found on chromosome 2 has two alleles – allele **A** for curly wing, which is dominant to allele **a** for normal wing. Female flies are homogametic while males are heterogametic. The gene for *Drosophila* eye colour is located on the X chromosome and has two alleles – the allele for red eyes being dominant to the allele for white eyes.

(a) (i) Write the genotypes of the two flies in the table below. [2]

	Fly 1	Fly 2
Sex of fly	Female	Male
Eye colour	White	Red
Wing shape	Curly wing, heterozygous	Curly wing, heterozygous
Genotype		

- (ii) Draw a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when these two flies are crossed. [4]

240 offspring were obtained from the above cross, and the phenotypes of the offspring were as follows:

83 red-eyed females with curly wings
 41 red-eyed females with normal wings
 78 white-eyed males with curly wings
 38 white-eyed males with normal wings

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

$$\chi^2 \text{ test: } \chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

Key to symbols: Σ = sum of ...
 v = degrees of freedom
 c = number of classes
 O = observed value
 E = expected value

Using the formula for the χ^2 test and the tables of χ^2 values above,

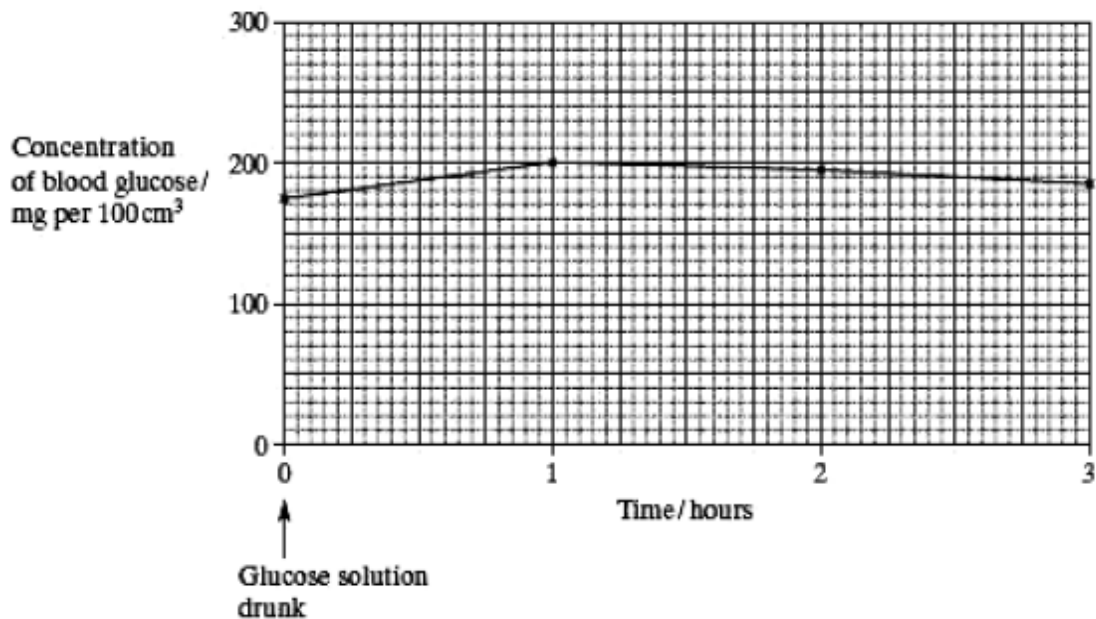
(b) (i) Calculate the value of the χ^2 test on these results. Show your working in the space provided below. [2]

(ii) State the degree of freedom for these results. [1]

(iii) Does this value for the chi-squared test support the expected phenotypic ratio that you have obtained in (a)(ii)? Explain. [3]

[Total: 12]

4. Some people produce no insulin. As a result, they suffer from a condition known as diabetes. During an investigation, a man with diabetes drank a glucose solution. The concentration of glucose in his blood was measured at regular intervals. The results are shown in the graph below.



- (a) Suggest one reason why the concentration of glucose decreased after 1 hour even though this man's blood contained no insulin. [1]

- (b) The investigation was repeated on a man who did not suffer from diabetes. The concentration of glucose on this man's blood before drinking the glucose solution was 80 mg per 100 cm³. Sketch a curve on the graph above to show the results you would expect. [1]

- (c) The diabetic man adopted a daily routine to stabilize his blood glucose concentration within narrow limits. He ate three meals a day: breakfast, a midday meal and an evening meal. He injected insulin once before breakfast and once before the evening meal.

The injection he used before breakfast was a mixture of two types of insulin - slow-acting insulin and fast-acting insulin.

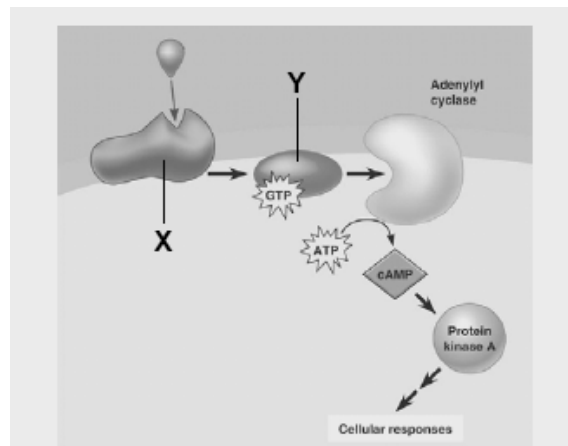
- (i) Explain the advantage of injecting both types of insulin before breakfast. [2]

- (ii) One day, the man did not eat a midday meal. Suggest one reason why his blood glucose concentration did not fall dangerously low even though he had injected himself with the mixture of insulin before breakfast. [1]

[Total: 5]

5. Certain microbes cause disease by disrupting the G-protein signaling pathways. The cholera bacterium, *Vibrio cholerae*, present in water contaminated with human waste colonizes the small intestine and produces a toxin that modifies a G protein.

The toxin results in a high concentration of cAMP which causes the intestinal cells to secrete large amounts of water and salt into the intestines, leading to profuse diarrhea and death from the loss of water and salts.



- (a) Name the molecules X and Y. [2]

X _____

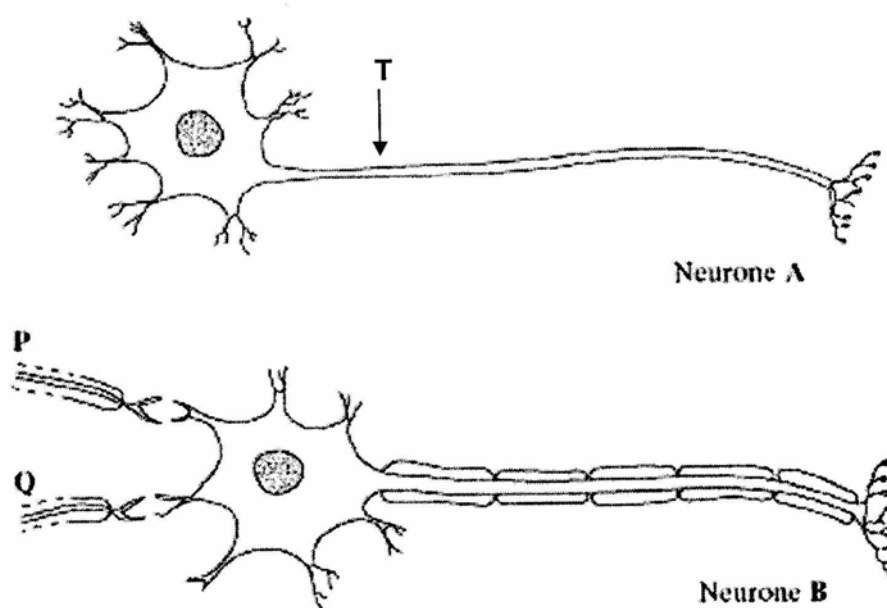
Y _____

- (b) What is the general name given to small molecules that function like cAMP in signaling pathways? [1]

- (c) Suggest how the toxin leads to an increase in cAMP concentration. [3]

[Total: 6]

6. The figure below shows two motor neurons, **A** and **B**. It also shows the synapses of neurone **B** with two other neurones, **P** and **Q**.



- (a) (i) If an action potential is initiated at point **T** on neurone **A**, describe how this action potential is propagated **along** the neurone. [3]

- (ii) Suggest why transmission of impulses along neurone **B** uses less energy than transmission along neurone **A**. [2]

- (b) GABA is a neurotransmitter present in some parts of the nervous system. The diagram below shows how the release of GABA from a presynaptic membrane affects the ion channels of a postsynaptic membrane.



- (i) Explain what would happen to the membrane potential on the postsynaptic membrane when GABA is released. [2]

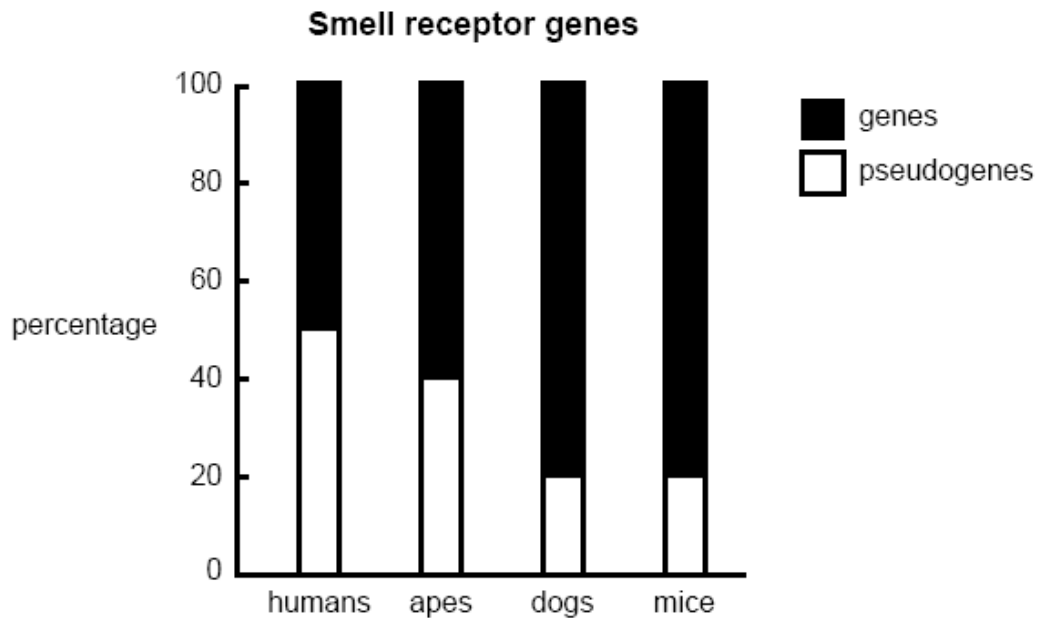
- (ii) Name the process whereby neurone **B** receives and integrates signals from neurones **P** and **Q** at the same time. [1]

- (iii) Explain why neurone **B** is less likely to respond if both neurones **P** and **Q** are stimulated at the same time but neurone **P** releases acetylcholine whereas neurone **Q** releases GABA. [2]

[Total: 10]

7. (a) Pseudogenes are the remains of broken genes which are unable to function and can be considered to be genetic fossils. Some are relics of genes lost through evolution while others reflect an earlier version of a present functional gene. Pseudogenes are able to accumulate all kinds of mutations.

In mammals, more than one thousand genes coding for smell receptors have been identified. Individual species vary in the proportion of these genes that have become pseudogenes, and humans have fewer than 500 functional genes coding for smell receptors. The majority of these genes are still functional genes in rats and mice.



- (i) From the graph, what can be concluded about the importance of smell to the survival of these four groups? [2]

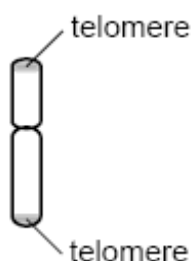
- (ii) Mice and dogs can only distinguish shades of grey, while some apes and monkeys are able to distinguish colours. Would you expect to find any change in the numbers of smell pseudogenes in apes and monkeys able to distinguish colours? Explain why. [2]

- (iii) Why are pseudogenes able to accumulate mutations that do not exist in functional genes? [2]

(b) Body cells have elaborate control systems that regulate their division. If these control systems break down, cells start to grow and divide in an unregulated way and produce a tumour.

(i) Cigarette smoking can lead to the formation of tumours. Describe how. [3]

Chromosomes are DNA structures in the nucleus of a cell. The ends of chromosomes are called telomeres.



Some changes in the DNA of these telomeres can trigger the development of cancer. Scientists are working to detect these changes in telomeres at an early stage of their development.

(ii) Outline one advantage of being able to detect early changes in telomere DNA. [1]

One normal sequence of DNA in a particular telomere was AATCGTCCAGGG. On examination of the telomeres of one patient, this sequence was found to have changed to AACCGTCCTGGG. The doctor examined the relationship between a DNA sequence and amino acids that could be expected. He used the following table to obtain information.

DNA sequence	Amino acid
AAT or AAC	leucine
CGT or CGG	alanine
CCA or CCT	glycine
GGG or GGT	proline

(iii) Is the doctor likely to be concerned or not concerned about the change in the telomere DNA of this patient? Explain your answer. [2]

[Total: 12]

8. Cytochrome c is found in the mitochondria of every aerobic eukaryote — animal, plant, and fungi. The amino acid sequences of cytochrome c from many of these organisms have been determined, and comparison made between the molecules shows that they are related.

Human cytochrome c contains 104 amino acids, and 37 of these have been found at equivalent positions in every cytochrome c that has been sequenced. **These molecules are homologous.**

Table 8 shows the first 18 amino acid residues of human cytochrome c, from the N-terminal, with the corresponding sequences from six other organisms aligned beneath. A ditto (“”) indicates that the amino acid is the same one found at that position in the human molecule. All the vertebrate cytochromes (the first four) start with glycine (Gly). The *Drosophila*, wheat, and yeast cytochromes have several amino acids that precede the sequence shown here (indicated by <<<).

Cytochrome	Amino acid																	
	1					6				10				14			17	18
Human	Gly	Asp	Val	Glu	Lys	Gly	Lys	Lys	Ile	Phe	Ile	Met	Lys	Cys	Ser	Gln	Cys	His
Pig	“	“	“	“	“	“	“	“	“	“	“	Val	Gln	“	“	Ala	“	“
Chicken	“	“	Ile	“	“	“	“	“	“	“	“	Val	Gln	“	“	“	“	“
Dogfish	“	“	“	“	“	“	“	“	“	Val	“	Val	Gln	“	“	Ala	“	“
<i>Drosophila</i>	<<<	“	“	“	“	“	“	“	“	Leu	“	Val	Gln	Arg	“	Ala	“	“
Wheat	<<<	“	Asn	Pro	Asp	Ala	“	Ala	“	“	“	Lys	Thr	“	“	Ala	“	“
Yeast	<<<	“	Ser	Ala	Lys	“	“	Ala	Thr	Leu	“	Lys	Thr	Arg	“	“	“	“

Table 8

- (a) With reference to Table 8, explain what is meant by “These molecules are homologous”. [3]

- (b) (i) Explain why the analysis of amino acid sequence is used to establish evolutionary relationship. [3]

- (ii) State a limitation of using difference in amino acid sequences of only one protein (cytochrome c) to establish phylogenetic relationship among the organisms. [1]

Cytochrome c is an ancient molecule which has coding and non-coding regions. It has evolved very slowly. Even after more than 2 billion years, one-third of its amino acids are unchanged. The conserved sequences are a great help in working out the evolutionary relationships between distantly-related organisms like fish and humans. However, for humans and the apes (e.g. chimpanzee), their cytochrome c molecules are identical and evolutionary relationships cannot be determined.

(c) State the name given to

Coding regions _____

Non-coding regions _____ [2]

(d) Suggest an alternative way by which the evolutionary relationship between humans and the apes can be determined by molecular techniques. [1]

[Total: 10]

Section B
Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections (a), (b) etc., as indicated in the question.

9. (a) Describe the structure of DNA and its arrangement in a chromosome. [8]
(b) Compare and contrast between DNA replication and transcription. [12]

OR

10. (a) Explain the importance of the light dependent reaction in photosynthesis. [7]
(b) Describe the role of NAD^+ in aerobic respiration. [6]
(c) Compare and contrast photophosphorylation and oxidative phosphorylation. [7]